SECTION 1

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. TOUGH: (A) neatly garbed (B) quietly stated (C) greatly improved (D) easily chewed (E) extremely clean

2. FEROCITY: (A) mildness (B) inaccuracy (C) originality (D) awkwardness (E) uselessness

3. ALLY: (A) miser (B) dictator (C) amateur (D) opponent (E) daredevil

4. GRACIOUS: (A) surly (B) vague (C) elderly (D) anonymous (E) miraculous

5. FIREBRAND: (A) genius (B) impostor (C) sentinel (D) spendthrift (E) peacemaker

6. GALL: (A) revive (B) murmur (C) soothe (D) prohibit disagreement (E) evaluate performance

7. FLAGRANT: (A) inferior (B) unnoticeable (C) infallible (D) conclusive (E) rancid

8. UNDULATING: (A) level (B) profuse (C) fragmentary (D) pompous (E) buoyant

9. EXTRICATE: (A) ripen (B) deceive (C) decrease (D) snare (E) invent

10. PLATITUDE: (A) valid criticism (B) dishonest approach (C) novel utterance (D) mammoth undertaking (E) significant progress

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. The castles of the feudal lords were primarily forts and therefore were constructed more for ---- than for ----.

(A) privacy..seclusion
(B) protection..comfort
(C) convenience..security
(D) permanence..preservation
(E) retaliation..defense

12. Those who ---- the use of force by labor unions should realize that management cannot be depended on to ---- a program of reform voluntarily.

(A) laud..initiate (B) ignore..discard (C) censure..deprecate (D) promote..effect (E) deplore..inaugurate

13. Standing up quietly for her rights, the heroine of Dickens' novel ---- bigots and ---- innumerable affronts to her personal dignity.

(A) denounced..replaced (B) converted..caused (C) trusted..expanded (D) shamed..weathered (E) insulted..inflicted

14. As if to ---- the towering buildings that now dominat our landscape, biographies of modern American authors depend on sheer ---- to compel authority.

(A) embody..intelligence (B) dismantle..ability (C) emulate..size (D) echo..talent (E) disregard..strength

15. Simmons could no longer be ----; with unusual ---- he castigated the council members for their persistent pettiness and provincialism.

(A) abusive..outrage (B) forbearing..venom (C) diplomatic..tact (D) polite..courtesy (E) censorious..scorn

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

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Princeton, N.J. 08541

1

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. MOTH : CLOTHING :: (A) woodpecker : hole
(B) bear : trap (C) lamb : wool
(D) puncture : tire (E) termite : house

17. FILTER : LIQUID :: (A) melt : ice
(B) grind : solid (C) sift : powder
(D) inflate : air (E) chisel : rock

18. ANNULMENT : MARRIAGE ::
(A) nullification : contract
(B) negation : opinion
(C) refutation : authority
(D) invalidation : discovery
(E) denial : permission

19. FRAME : PAINTING :: (A) orchestra : violin
(B) shore : lake (C) stage : balcony
(D) window : wall (E) lock : cell

20. HIJACK : GOODS :: (A) embezzle : bank
(B) slander : reputation (C) trespass : boundary
(D) forge : document (E) kidnap : person

21. CONSERVATORY : MUSIC :: (A) anthology : books
(B) aerie : birds (C) bivouac : army
(D) seminary : religion (E) arbor : grapes

22. HOMOGENEOUS : KIND :: (A) contemporary : age
(B) enigmatic : force (C) precipitous : length
(D) superficial : surface (E) suitable : form

23. MAGICIAN : PRESTIDIGITATION ::
(A) masquerader : conspiracy
(B) matador : peccadillo
(C) gymnast : omnipotence
(D) clown : pugilism
(E) mimic : imitation

24. PRATTLE : DISCOURSE :: (A) trinket : treasure
(B) epithet : effigy (C) halo : crown
(D) shot : volley (E) answer : response

25. DOUBLE-DEALER : DUPLICITY ::
(A) profligate : dissipation
(B) insurgent : oppression
(C) perjurer : negligence
(D) counterfeiter : authenticity
(E) authoritarian : disobedience

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Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

(The passages for this test have been adapted from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service.)

In past days Captain Bailey had handled many thousands of pounds of his employers' money and of his own; he had attended faithfully, as by law a shipmaster is expected to do, to the conflicting interests of owners, charterers, and underwriters. He had never lost a ship or consented to a shady transaction. He had buried his wife, had married off his daughter to a man of her unlucky choice, and had lost much in the crash of the notorious Travanacore and Deccan Banking Corporation, whose downfall had shaken the East like an earthquake. And he was sixty-seven years old.

His age sat lightly on him, and of his ruin he was not ashamed. He had not been alone in believing in the stability of the Banking Corporation. Men whose judgment in matters of finance was as expert as his seamanship had commended the prudence of his investments, and had themselves lost much money in the great failure. The only difference between him and them was that he had lost his all and yet not his all. There remained to him from his lost fortune a very pretty little barque, Fair Maid, which he had bought to occupy his leisure as a retired sailor.

He had formally declared himself tired of the sea the year preceding his daughter's marriage. But after the young couple had gone to settle in Melbourne, he found that he could not make himself happy on shore. He was too much of a merchant sea-captain for mere yachting to satisfy him. He wanted the illusion of affairs, and his acquisition of Fair Maid preserved continuity in his life. He introduced her to his acquaintances in various ports as "my last command." When he grew too old to be trusted with a ship, he would lay her up and go ashore and be buried, leaving directions in his will to have the barque towed out and scuttled on the day of the funeral. His daughter would not begrudge him the satisfaction of knowing that no stranger would handle his last command after him. All this would be said with a jocular twinkle in his eye: the vigorous old man had too much vitality for the sentimentalism of regret; and yet a little wistfully because he was at home in life, taking genuine pleasure in its feelings and possessions, in the dignity of his reputation, in his love for his daughter, and in his satisfaction with the ship—the plaything of his lonely leisure.

26. The passage primarily concerns

(A) Captain Bailey's memories of his career at sea
(B) the events that contributed to Captain Bailey's success as a shipmaster
(C) Captain Bailey's actions and attitudes after retirement
(D) the conditions that caused the loss of Captain Bailey's fortune
(E) the experiences shared by Captain Bailey and Fair Maid

27. The expression "his ruin" (line 13) refers to Captain Bailey's

(A) physical condition
(B) discredited reputation
(C) loss of his fortune
(D) isolation from his former friends
(E) departure from his job as shipmaster

28. It can be inferred that Captain Bailey's employers believed that, as a shipmaster, he was

(A) loyal but unambitious
(B) competent and ethical
(C) talented but undependable
(D) eccentric and unpopular
(E) restless and erratic

29. Captain Bailey wished to have Fair Maid sunk on the day of his funeral because he

(A) knew that no one else could handle the boat
(B) resented the boat's seductive power over him
(C) realized that maintaining the boat would be a burden for his daughter
(D) did not want the boat to be under any command but his own
(E) feared that the boat would fall into the hands of his creditors

30. Captain Bailey regarded Fair Maid with

(A) weary resignation
(B) casual indifference
(C) sentimental regret
(D) awe-struck fascination
(E) affectionate delight

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1

Any survey of medieval town life delights in the color of guild organizations: the broiders and glovers, the shipwrights and upholsters, each with its guild hall, its distinctive livery, and its elaborate set of rules. But if life in the guilds and at the fairs provides a sharp contrast with the stodgy life on the manor, we must not be misled by surface resemblances into thinking that guild life represented a foretaste of modern life in medieval dress. It is a long distance from guilds to modern business firms, and it is well to fix in mind some of the differences.

In the first place, the guild was much more than just an institution for organizing production. Whereas most of its regulations concerned wages and conditions of work and specifications of output, they also dwelt at length on noneconomic matters: on a member's civic role, on his appropriate dress, and even on his daily deportment. Guilds were the regulators not only of production but of social conduct.

Between guilds and modern business firms there is a profound gulf. Unlike modern firms, the purpose of guilds was not first and foremost to make money. Rather, it was to preserve a certain orderly way of life—a way which envisaged a decent income for the master craftsmen but which was certainly not intended to allow any of them to become "big" businessmen. On the contrary, guilds were specifically designed to ward off any such outcome of an uninhibited struggle among their members. The terms of service and wages were fixed by custom. So, too, were the terms of sale: a guild member who cornered the supply of an item or bought wholesale to sell at retail was severely punished. Competition was strictly limited and profits were held to prescribed levels. Advertising was forbidden, and even technical progress in advance of one's fellow guildsmen was considered disloyal.

Surely the guilds represent a more "modern" aspect of feudal life than the manor, but the whole temper of guild life was still far removed from the goals and ideals of modern business enterprise. There was no free competition and no restless probing for advantage. Existing on the margin of a relatively moneyless society, the guilds were organizations that sought to take the risks out of their slender enterprises. As such, they were as drenched in the medieval atmosphere as the manors.

31. The author is primarily concerned with

(A) analyzing the origins of the guild system
(B) explaining the relationships between manors, fairs, and modern business firms
(C) depicting the weaknesses of the guilds' business practices
(D) stressing the historical evolution of guilds to modern business firms
(E) discussing some differences between medieval and modern business practices

32. According to the passage, which of the following would LEAST likely be found in a guild handbook?

(A) The fees a master guildsman should charge
(B) The bonus a member would receive for record sales
(C) The maximum number of hours a guildsman would be expected to work
(D) The steps a new shipwright would follow to become a master craftsman
(E) The organizations to which a member should contribute as an upstanding citizen

33. According to the passage, modern business enterprises, compared to the medieval guilds, are

(A) more concerned with increasing profits
(B) influenced more by craftsmen than by tradesmen
(C) more subordinate to the demands of consumers
(D) less progressive in financial dealings
(E) less interested in quantity than quality

34. It can be inferred that the guilds were organized as they were because

(A) life on the manors was boring and drab
(B) technical improvements were still improbable
(C) they stressed preservation and stability, not progress
(D) people in medieval times were interested in advancing individual liberty
(E) social status was determined by income

35. With which of the following statements concerning modern business firms would the author be most likely to agree?

(A) They make rules concerning appropriate business practices for employees.
(B) They permit the free play of price in terms of service and sales.
(C) Their main concern is the stability of profit levels.
(D) Their aim is to discourage competition among independent manufacturers.
(E) They are organized in such a way that co-operating monopolies will develop.

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All migrating birds must guide their flight by means some aspect of their environment which is related in reasonably reliable fashion to the direction of the goal. e must also take as certain, unless we are to fall back extrascientific theories, that this environmental cue ust be perceived by the birds; it must stimulate some ense organ or receptor cells, for these are the only nctional contacts between a bird's nervous system and s outside environment. Our problem, then, is to find e environmental cue, and also to find the sensory echanism by which this cue is recognized and chaneled into the central nervous system, where it can esult in the appropriate actions to move the bird in the ght direction.

Some investigators have tried to account for bird avigation in terms of the known sensory mechanisms, hich are the same in all higher vertebrates. For xample, some have suggested that migrants are guided visual landmarks. But the overwater routes such as ose of the golden plover are devoid of topography for ndreds of miles. Others feel that wind direction may fer a guiding cue, but winds are notoriously changeble. Only if the bird knew the wind direction characteristic of each type of air mass and weather condition ong its route could it guide itself from the arctic to e tropics.

On the other side, we find a variety of ingenious eories hung up on formidable objections from the oint of view of the sensory physiologist. Best known, erhaps, are a number of theories based on the idea at birds have the equivalent of a magnetic compass—at they can perceive the earth's magnetic field in me manner and guide their migrations accordingly. hose who subscribe to these theories must face the ct, however, that no one has been able to show that rds can sense a magnetic field as weak as the earth's, y more than we ourselves can.

A really new theory was advanced recently, namely at birds orient themselves by means of mechanical rces arising from the earth's rotation. One of the rms such forces may take is an increase in the parent weight of a flying bird depending on the direcon of its flight. This effect is of a type which might within the range of a bird's sense organs, since it volves mechanical acceleration for which the bird s specialized receptors in the inner ear labyrinth. ut the difficulties in the way of the bird's being able make the necessary quantitative distinctions are normous. The variation in weight is only one part in everal thousand, and it could easily be masked by the uch larger accelerations resulting from the flight self, to say nothing of the slightest turbulence of the r, or even the bird's own breathing and heartbeat. hus this hypothesis seems scarcely more plausible an the magnetic theories.

36. Which of the following best describes the passage?

(A) A comprehensive account of each of the theories that has been developed to explain bird navigation
(B) A description of a theory of bird navigation followed by specific supporting evidence
(C) An account of the different methods of navigation used by different varieties of birds
(D) A statement of the general problem of explaining bird navigation followed by examples of attempted solutions
(E) An account of the findings of research in bird navigation conducted by the author

37. Which of the following is an assumption made by the author?

(A) Most migrating birds navigate in the same manner.
(B) The sensory mechanism used by birds in navigation is common to all higher vertebrates.
(C) Migrating birds can fly in the correct direction without aid from environmental clues.
(D) Birds have more highly developed sense organs than most other animals.
(E) Birds have no specific goal in mind when they begin a migration.

38. Which of the following best states the author's opinion of the theories discussed in the last two paragraphs?

(A) The theories are not themselves complete explanations of bird navigation, but they are steps in the right direction.
(B) The theories are essentially extrascientific and not worthy of the attention of serious scientists.
(C) The theories are very clever and interesting, but they are probably untrue.
(D) The theories are untrue, since they are contradicted by the known facts of bird physiology.
(E) The theories are probably true, but they do not really explain bird navigation.

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39. The author's objection to the theory that birds navigate by means of forces arising from the earth's rotation is based primarily on the fact that

(A) the environmental cue is weak relative to other forces that affect the bird in a similar manner
(B) the receptors through which the bird perceives mechanical acceleration are not sufficiently sensitive
(C) a bird is not heavy enough to be affected by the earth's rotation
(D) the effects of the earth's rotation on a bird are not related to direction of flight
(E) birds are not intelligent enough to determine their direction relative to the earth's rotation

40. The author's objection to the "magnetic" theory of bird navigation is based on which of the following?

I. The environmental cue is not related reliably to the direction of the goal.
II. There is no known sensory mechanism by which the environmental cue can be perceived.
III. The weak effects of the earth's magnetic field are masked by other forces acting on the bird.

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) II and III only

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

35 QUESTIONS

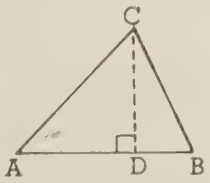
this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

he following information is for your reference in solving some of the problems.

Circle of radius r: Area = πr^2; Circumference = $2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

= is equal to	≦ is less than or equal to
≠ is unequal to	≧ is greater than or equal to
< is less than	∥ is parallel to
> is greater than	⊥ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180

If ∠CDA is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

ote: Figures which accompany problems in this test are intended to provide information useful in solving the problems. hey are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to cale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} =$

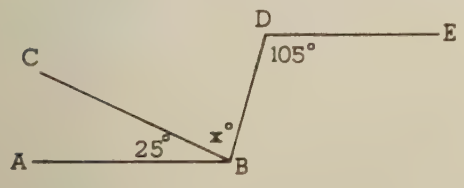
(A) $\frac{1}{4}$ (B) $\frac{3}{5}$ (C) $\frac{10}{7}$ (D) $\frac{17}{10}$ (E) $\frac{23}{12}$

2. If $N \div 6 = K$, which of the following statements must be true?

(A) $K \div 6 = N$
(B) $6 \div K = N$
(C) $6 \times N = K$
(D) $6 \times K = N$
(E) $N \times K = 6$

3. If $\frac{2}{3}$ the perimeter of an equilateral triangle is 6, what is its perimeter?

(A) 4 (B) 8 (C) 9 (D) 12 (E) 18



4. In the figure above, if AB is parallel to DE, then x =

(A) 105 (B) 90 (C) 80 (D) 75 (E) 65

5. If the sum of $t - 1$, $t - 2$, $t - 3$ is 0, then $t =$

(A) −2 (B) −1 (C) 0 (D) 1 (E) 2

6. In 1969 a seaport handled 120,000 tons of cargo. If it handled 180,000 tons in 1973, the average annual increase in tons handled was

(A) 12,000
(B) 15,000
(C) 18,000
(D) 20,000
(E) 45,000

7. If p, q, r, s, and t are different numbers and each is equal to one of the numbers 1, 2, 4, 8, and 16 and if $4p = \frac{1}{2}q = 8r = s = 2t$, then $q =$

(A) 1
(B) 2
(C) 4
(D) 8
(E) 16

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2

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. A symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	$180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$e(f + e)$	$(e + f)e$

Three years ago, Gina was 2 centimeters taller than Tina.

	Column A	Column B
9.	Tina's height today	Gina's height today

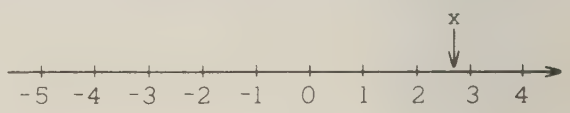
25% of $x = 60$

	Column A	Column B
10.	50% of x	120

	Column A	Column B
11.	The average (arithmetic mean) of x and y	x

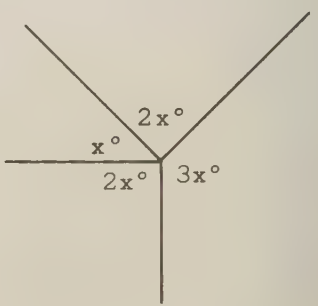
	Column A	Column B
12.	The number of odd integers greater than 2 and less than 10	The number of prime integers greater than 2 and less than 10

	Column A	Column B
13.	Sum of any two negative numbers	Sum of any two positive numbers



	Column A	Column B
14.	$2 - x$	0

	Column A	Column B
15.	The greatest integer x such that $x^2 < 65$	The greatest integer y such that $y^3 < 65$



	Column A	Column B
16.	x	45

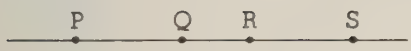
$$\begin{cases} x + 2y = 10 \\ 2x - 2y = 5 \end{cases}$$

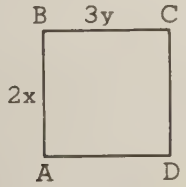
	Column A	Column B
17.	x	y

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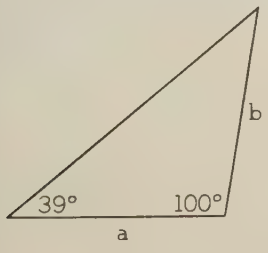
SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Column A	Column B
 PQ > RS	
PR	QS

Column A	Column B
 ABCD is a square.	
$4x^2$	$9y^2$

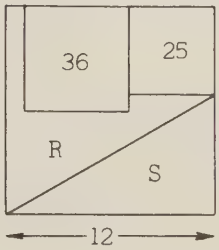
Column A	Column B
$s \neq 0$	
$\frac{r}{s}$	rs

Column A	Column B
	
a	b

Column A	Column B
$2x < 4$ $3y < 9$	
x	y

Column A	Column B
Let n* be defined by the equation $n^* = (n-1)(n)(n+1)$ where n is an integer.	
$\frac{6^*}{3^*}$	2^*

	Column A	Column B
24.	The square of $\frac{13}{19}$	The square root of $\frac{13}{19}$



The square with side 12 above is divided into squares with areas 36 and 25 and two nonoverlapping regions with areas R and S.

	Column A	Column B
25.	R	S

$\frac{1}{2}, \frac{2}{3}, \frac{3}{5}, \frac{5}{8}, \frac{8}{13}$

	Column A	Column B
26.	The greatest fraction shown	The product of the least fraction shown and $\frac{4}{3}$

$x - y = 5$

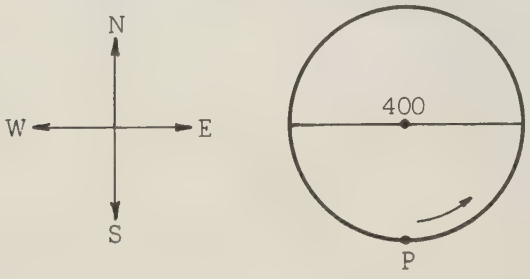
	Column A	Column B
27.	$\frac{1}{x} - \frac{1}{y}$	$\frac{1}{5}$

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then indicate the on correct answer in the appropriate space on the answer sheet.

28. If $\frac{k}{2}$ is an even number and $\frac{k}{4}$ is an odd number, then k could equal

(A) 24 (B) 20 (C) 16 (D) 10 (E) 8



29. A circular track 400 meters in diameter is shown in the figure above. A runner starts at P, directly south of the center of the track, and runs counter-clockwise. At the end of exactly how many meters of travel will the runner be at the point where he is traveling directly north?

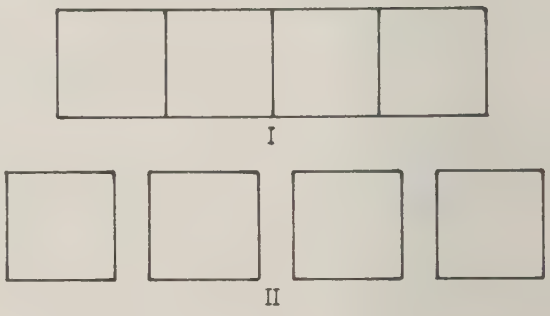
(A) 25π (B) 100π (C) 400π
(D) 800π (E) $1{,}000\pi$

30. If x ranges in value from 0.0001 to 0.01 and y ranges in value from 0.001 to 0.1, what is the maximum value of $\frac{x}{y}$?

(A) 0.001 (B) 0.1 (C) 1
(D) 10 (E) 1,000

31. Jane owes Bill \$20, Bill owes George \$30, and George owes Jane \$40. All of these debts may be settled if

(A) Bill and George each pay Jane \$10
(B) Bill pays George \$10 and George pays Jane \$10
(C) George pays Jane and Bill \$10 each
(D) George pays Jane \$20
(E) Jane pays George \$20



32. All of the squares above are equal in area; the to area of the four squares in each figure is 36. T total length of the line segments that form figure is how much greater than the total length of those that form figure I ?

(A) 0 (B) 3 (C) 6 (D) 9 (E) 12

33. If $\left(x + \frac{1}{x}\right)^2 = 25$, then $\frac{1}{x^2} + x^2 =$

(A) 23
(B) 24
(C) 25
(D) 27
(E) 624

34. What is the maximum number of points that can be located within or on the sides of a square of side 2 if the distance between any two of the points is to be greater than or equal to $\sqrt{2}$?

(A) Three (B) Four (C) Five
(D) Six (E) Seven

35. An agency rents automobiles at a rate of d dollar per day and x cents per kilometer. If a car was rented for 24 days and the odometer showed an addition of 1800 kilometers traveled, then the amount, in dollars, due the agency was

(A) $\frac{24(d + 75x)}{100}$ (B) $\frac{6(3d + 400x)}{100}$ (C) $24(d +$

(D) $6(3d + 4x)$ (E) $6(4d + 3x)$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 4

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. DISTRIBUTE: (A) unveil (B) repay (C) collect (D) burden (E) refute

2. GRATIFYING: (A) inexpensive (B) unsatisfying (C) stultifying (D) unappetizing (E) pessimistic

3. HIDEOUS: (A) timid (B) genuine (C) fragile (D) obvious (E) lovely

4. FLOWER: (A) wither (B) envelop (C) soften (D) simplify (E) challenge

5. BOISTEROUS: (A) courageous (B) skillful (C) tolerant (D) serene (E) supernatural

6. GOUGE: (A) hold up (B) fill in (C) eat slowly (D) disengage (E) mount

7. HILARITY: (A) rivalry (B) obscurity (C) cleverness (D) viciousness (E) downheartedness

8. CORRUGATED: (A) distended (B) refined (C) uninsulated (D) smooth (E) honest

9. CIRCUITOUS: (A) involuntary (B) straightforward (C) lily-livered (D) dog-eared (E) homely

10. COALESCE: (A) disrupt (B) disprove (C) disperse (D) disown (E) dislike

11. LEVITY: (A) seriousness (B) indifference (C) purity (D) uniformity (E) incoherence

12. COGNIZANT: (A) practical (B) ignorant (C) hostile (D) unattractive (E) silent

13. DEARTH: (A) power (B) liveliness (C) serious attempt (D) adequate supply (E) persuasive personality

14. UNWITTING: (A) intelligent (B) conscious (C) disappointing (D) overly curious (E) in plain sight

15. ASSUAGE: (A) intensify (B) promote (C) trust (D) mislead (E) isolate

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. No longer is it enough to ---- our present resources; somehow we must find ways to replace what we have lost.

(A) squander (B) conserve (C) belittle (D) eliminate (E) utilize

17. Stability in a culture does not mean inertia; even the most ---- culture is marked by constant ----.

(A) mobile..inflexibility
(B) materialistic..uncertainty
(C) progressive..failure
(D) advanced..variety
(E) conservative..change

18. Williams rather likes the present tenure policy and would be distressed if it were ----.

(A) emulated (B) tolerated (C) nurtured (D) extolled (E) superseded

19. Crawford is unnecessarily ---- in assessing the views of others; he often describes the conclusions of respected scholars as "not worthy of being refuted."

(A) contrite (B) verbose (C) disparaging (D) diligent (E) hesitant

20. You have only to study the intricate thoughts of the great philosophers to see that it is possible to express with ---- the most ---- reflections.

(A) incisiveness..clear-cut
(B) refinement..pleasant
(C) unanimity..disagreeable
(D) emotion..banal
(E) lucidity..subtle

GO ON TO THE NEXT PAGE

4

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basi of what is stated or implied in that passage.

(The passages for this test have been adapted from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service.)

William H. Johnson's best paintings, which express great warmth and humanity, represent a significant contribution by a black American to modern art. Johnson has been described, though somewhat inadequately, as an American primitive painter. Unlike the true primitive, Johnson deliberately chose a naïve approach; his relationship to naïve art is similar to the relationship of some expressionists to African sculpture. He borrows the forms in order to express new ideas.

Johnson spent his early years in Scandinavia painting impressionistic landscapes, which are technically competent but of little artistic interest. However, it is in his later religious art, using primitive techniques, that William H. Johnson finds his true expression. His pictures from the life of Christ embody the simplicity and sincerity of the Negro spiritual and are extremely moving. The secular pictures of this primitive period, however, usually fail to achieve any emotional depth. His "Two Sisters," for example, is a decorative, attractive picture, but the artist does not transcend the primitive medium he has deliberately chosen. Again, in the "Funeral," the self-conscious primitivism is obvious and awkward. The facial expressions are empty; the large gestures are melodramatic and somewhat posterlike.

In his series of works from the life of Christ, Johnson develops a far greater sensitivity. How infinitely delicate is his drawing of the baby Christ in his "Nativity"; how movingly sincere and pious the simple gestures of the parents. Here the naïve approach is used to good purpose. A more sophisticated approach to perspective, anatomy, texture, light and shade, etc., would have detracted from the pure, simple emotion the artist wants to portray.

Johnson's treatment of hands is extremely expressive. They are usually very large, and sometimes, as in Christ's hands in "Let Little Children Come unto Me," their ecstatic gestures dominate the entire composition. In the "Entombment," his treatment of feet is equally expressive. The feet here not only produce the gentle rhythm and movement of the picture but also express sorrow. Johnson uses his particular forms of distortion and simplification to express simplicity and sincerity, and he also achieves a feeling of great intimacy.

Occasionally he produces works of unusual intensity. His "Crucifixion I" is a picture of great power. In its heightened expression of pain, particularly in the contorted pain of Mary, the picture is reminiscent of Grünewald's famous Isenheim altar.

21. It can be inferred that Johnson's early impressionistic work was

(A) artistically undistinguished
(B) stylistically innovative
(C) emotionally sensitive
(D) generally underrated
(E) belatedly popular

22. The author is mainly concerned with Johnson's

(A) life story (B) artistic style
(C) reasons for painting (D) religious beliefs
(E) technical training

23. According to the passage, Johnson's treatment of feet expressed sorrow in which of the following works?

(A) "Funeral"
(B) "Nativity"
(C) "Let Little Children Come unto Me"
(D) "Entombment"
(E) "Crucifixion I"

24. The author describes the gestures depicted in whic of the following paintings as "posterlike" (line 27)

(A) "Two Sisters" (B) "Funeral" (C) "Nativi
(D) "Entombment" (E) "Crucifixion I"

25. It can be inferred that the author finds the descrip tion of Johnson as an American primitive to be son what inadequate because Johnson has

(A) produced abstract paintings, whereas Americ primitive art is generally representational
(B) built his reputation as an impressionistic painter
(C) used primitive techniques only in his very early works
(D) produced profane pictures, whereas American primitive painting is entirely religious
(E) expressed new ideas by means of the primitive artistic approach

GO ON TO THE NEXT PAGE

The people of the United States have the undoubted right to change their form of government and to renounce established customs or longstanding policies whenever in their wisdom they see fit to do so. As a believer in democratic government, I readily acknowledge the right of the people to make in an orderly fashion such changes as may be approved in their judgment at any time. I contend, moreover, that when radical and important departures from established national policies are proposed, the people ought to be consulted.

There has been proposed what to my mind is the most radical departure from our policies that has ever been proposed at any time since our government was established. I think even the advocates of this program agree with me that it is a pronounced departure from all the policies we have heretofore obtained. It may be wise, as they contend; nevertheless, it involves a different course of conduct on the part of the government and of our people for the future, and the people are entitled to pass judgment on the advisability of such a course.

Certainly questions of this kind ought to be submitted to a plebiscite, a vote of the people. We are merely agents of the people, and it will not be contended that we have received any authority from the principal, the people, to proceed along this line. It is a greater responsibility than an agent ought to assume without express authority or approval from his principal. Preliminary to a discussion of this question, therefore, I want to declare my belief that we should arrange the machinery for taking a vote of the people of the United States upon this stupendous program. I am aware that the processes by which that may be accomplished involve some difficulties; but they are not insurmountable, and they are by no means to be compared in their difficulty with the importance of being right and in harmony with the judgment of the people before we proceed to the final approval. We should have the specific endorsement of those whose agents we are for the fearful responsibility we propose to assume. If we can effectuate this change now proposed without direct authority from the people, I cannot think of a question of sufficient moment to call for their endorsement.

26. The author's primary purpose is apparently to

(A) persuade citizens to make full use of their voting privileges
(B) suggest needed changes in the manner of voting for government officials
(C) deter legislators from making unnecessary changes in national policies
(D) convince the public that a proposed program is detrimental to the nation
(E) urge that the public be consulted before radical changes in national policy are made

27. The author implies that those who believe that the proposed program involves substantial changes in national policy include

(A) only the conservative legislators
(B) only the opponents of the program
(C) only the advocates of the program
(D) both the opponents and the advocates of the program
(E) both the legislators and the populace

28. It can be inferred that the author's concern with the proposed program is that of

(A) an irate citizen
(B) an uninformed commentator
(C) a wary opponent
(D) an approving legislator
(E) a bitter and critical historian

29. Which of the following best describes the author's attitude in the last sentence of the passage?

(A) Enthusiastic optimism
(B) Sympathetic encouragement
(C) Momentary doubt
(D) Solemn apprehension
(E) Flippant sarcasm

30. According to the author, the effort of taking a nation-wide vote on the proposed program would be

(A) meaningless and unsuccessful
(B) necessary and uncomplicated
(C) advisable but impossible
(D) arduous but worthwhile
(E) unwieldy and futile

GO ON TO THE NEXT PAGE

4

Select the word or set of words that best completes each of the following sentences.

31. This probing and startling analysis challenges us to ---- our ---- attitudes and look at the world in a totally new way.

(A) defend..immutable
(B) discard..preconceived
(C) shelve..novel
(D) affirm..conventional
(E) waive..innovative

32. Many people believe that malnutrition and its ill effects occur only in developing countries, but the fact is that our own population is far from ----.

(A) healthy (B) informed (C) nutritious
(D) susceptible (E) impoverished

33. The admiral's tolerance of ---- among the officers was ---- by his desire for loyalty from all.

(A) flattery..offset
(B) dissent..tempered
(C) compliance..undermined
(D) defiance..intensified
(E) discipline..corrupted

34. Hinckle tried to explain the differences between those controversies that demand an absolutist, moralistic approach and those that call for a pragmatic, ---- solution.

(A) conciliatory (B) permanent (C) immoderate
(D) categorical (E) inexpedient

35. Generally ---- the opportunity to join the gentry and live the aristocratic life, the British scientists of this period brought to their laboratories a devotion to their calling akin in its intensity to ---- asceticism.

(A) pursuing..medieval
(B) exploiting..orthodox
(C) eschewing..monastic
(D) coveting..intellectual
(E) flaunting..hypocritical

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pai

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. INTERMISSION : PERFORMANCE ::
(A) break : work (B) backlog : shipping
(C) solo : melody (D) union : strike
(E) skirmish : fighting

37. SNIP : SCISSORS :: (A) chop : ax
(B) darn : socks (C) flow : river
(D) pedal : bicycle (E) puff : pipe

38. TITTER : LAUGHTER :: (A) patter : stumbling
(B) paddle : canoeing (C) flipper : swimming
(D) ladder : height (E) whisper : speech

39. QUARREL : DEBATE :: (A) marathon : victory
(B) invasion : compromise (C) brawl : duel
(D) riot : defeat (E) permit : license

40. POACHER : GAME :: (A) sentry : fort
(B) archer : quiver (C) fugitive : clue
(D) swindler : truth (E) pickpocket : wallet

41. UNIQUE : DUPLICATE :: (A) substantial : base
(B) resonant : tone (C) abstract : merit
(D) eternal : end (E) poetic : rhythm

42. LEAN : EMACIATED :: (A) weary : exhausted
(B) large : expanded (C) ill : cured
(D) swollen : shrunken (E) narrow : enclosed

43. PREEMINENT : WORTH :: (A) eccentric : popularity
(B) insolvent : funds (C) complacent : efficiency
(D) optimistic : wealth (E) noble : character

44. TACITURN : CHATTER ::
(A) clever : think
(B) ascetic : indulge
(C) benevolent : donate
(D) authoritative : administer
(E) inquisitive : inquire

45. FLUSTERED : COMPOSURE ::
(A) lukewarm : enthusiasm (B) jealous : elegance
(C) sluggish : endurance (D) boring : patience
(E) aggressive : activity

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

25 QUESTIONS

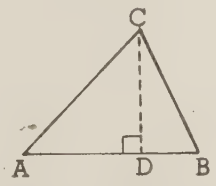
this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

he following information is for your reference in solving some of the problems.

Circle of radius r: Area = πr^2; Circumference = $2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

=	is equal to	≦	is less than or equal to
≠	is unequal to	≧	is greater than or equal to
<	is less than	∥	is parallel to
>	is greater than	⊥	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

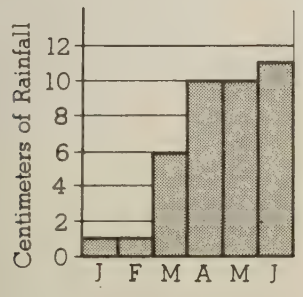
(2) $AC^2 = AD^2 + DC^2$

ote: Figures which accompany problems in this test are intended to provide information useful in solving the problems. hey are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to cale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $11 + xy = 12 + 11$, then the pair x, y could be any of the following EXCEPT

(A) 12, 1
(B) 6, 6
(C) 4, 3
(D) 3, 4
(E) 2, 6



2. The graph above shows the monthly rainfall for a six-month period in a region of Ghana. What was the total rainfall in centimeters for this period?

(A) 28
(B) 29
(C) 38
(D) 39
(E) 40

3. A woman sailed 1 kilometer east and then 1 kilometer south. In what direction would she have to go to return directly to her starting point?

(A) North (B) Northeast (C) Northwest
(D) Southeast (E) Southwest

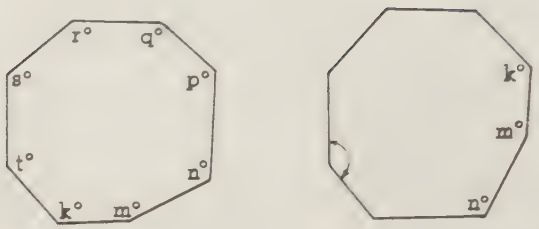
4. If the measure of one acute angle of a right triangle is 35°, then the measure of the other acute angle is

(A) 35° (B) 55° (C) 65° (D) 90° (E) 145°

5. The formula for the power P of an electric appliance is $P = EI$ where E is the number of volts and I the number of amperes of current. What is I when $P = 1{,}620$ and $E = 120$?

(A) 13.5
(B) 19.4
(C) 135
(D) 19,440
(E) 194,400

GO ON TO THE NEXT PAGE



6. If the octagon on the left above is turned over and placed in the position indicated by the angle measures in the octagon on the right, what is the measure of the marked angle?

(A) p° (B) t° (C) q° (D) s° (E) r°

---

7. The following are the dimensions of five boxes. All have equal volume EXCEPT the one that is

(A) 8 by 3 by 5
(B) 2 by 12 by 5
(C) 2 by 3 by 20
(D) 4 by 6 by 5
(E) 3 by 4 by 14

8. Which of the following conditions will make $x - y$ a positive number?

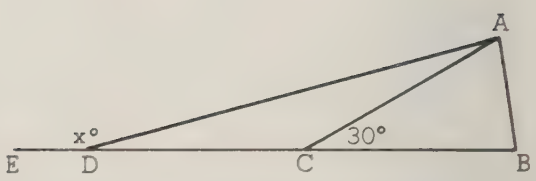
(A) $0 < y$
(B) $y < x$
(C) $x < 0$
(D) $x < y$
(E) $x = y$

9. What is half of the perimeter of a square of area 36 ?

(A) 3 (B) 6 (C) 12 (D) 18 (E) 24

10. If $x(3y) = (5)(3)$, then $xy - 1 =$

(A) 4 (B) 5 (C) 14 (D) 15 (E) 39



11. In the figure above, if $AC = CD$, then $x =$

(A) 105
(B) 120
(C) 135
(D) 150
(E) 165

---

12. If $abe = 0$, $bcd = 1$, and $bce = 0$, which of the following must equal 0 ?

(A) a (B) b (C) c (D) d (E) e

13. Container R is full of water and has twice the volume of container S, which is empty. If $\frac{1}{4}$ of the water in R is poured into S, what fraction of the volume of S is occupied by water?

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

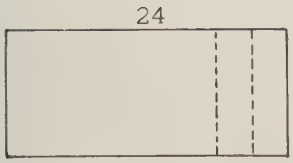
(E) None of the above

2	X	14	Y
X	14	Y	2
14	Y	2	X
Y	2	X	14

14. In the figure above, the sums of the numbers in each row, column, and main diagonal are the same. What is the value of X ?

(A) 2
(B) 8
(C) 12
(D) 14
(E) 16

GO ON TO THE NEXT PAGE



15. In the figure above, a rectangular piece of paper with length 24 is cut into strips of equal width as indicated by the dotted lines. If the area of two of these strips is 25 per cent of the area of the rectangle, how wide is each strip?

(A) 8 (B) 6 (C) 4 (D) 3 (E) 2

---

16. If $3x + 2y + xy + 6 = 0$ and $x + 2 = 5$, then $3 + y =$

(A) −15 (B) −3 (C) 0 (D) 3 (E) 15

17. P, Q, R, S, and T are five distinct lines in a plane. If $P \perp Q$, $Q \perp R$, $S \perp T$, and $R \parallel S$, then all of the following are true EXCEPT

(A) $P \parallel R$ (B) $P \parallel S$ (C) $P \perp T$
(D) $S \perp Q$ (E) $Q \perp T$

18. What is the difference between the greatest and least of all three-digit positive integers, each of whose digits is a different nonzero multiple of 3 ?

(A) 324 (B) 540 (C) 567 (D) 594 (E) 604

Questions 19-20 refer to the following definition.

$\left.\begin{array}{|c|c|c|}\hline a & b & c \\ \hline\end{array}\right. = \dfrac{\frac{a}{b}}{c}$ for all nonzero real numbers b and c.

19. $\left.\begin{array}{|c|c|c|}\hline 3 & 5 & 6 \\ \hline\end{array}\right. =$

(A) $\frac{1}{10}$ (B) $\frac{1}{6}$ (C) $\frac{1}{5}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

20. $\left.\begin{array}{|c|c|c|}\hline 8 & 16 & 8 \\ \hline\end{array}\right.$ is equal to which of the following?

(A) $\left.\begin{array}{|c|c|c|}\hline 1 & 2 & 1 \\ \hline\end{array}\right.$ (B) $\left.\begin{array}{|c|c|c|}\hline 2 & 16 & 2 \\ \hline\end{array}\right.$ (C) $\left.\begin{array}{|c|c|c|}\hline 8 & 2 & 8 \\ \hline\end{array}\right.$

(D) $\left.\begin{array}{|c|c|c|}\hline 16 & 8 & 16 \\ \hline\end{array}\right.$ (E) $\left.\begin{array}{|c|c|c|}\hline 4 & 1 & 1 \\ \hline\end{array}\right.$

---

21. Three line segments have lengths $2x - 1$, $x + 2$, and $3x - 4$, respectively. If the average length of the line segments is 9, how much longer is the longest segment than the shortest?

(A) 3
(B) 4
(C) 5
(D) 7
(E) 11

GO ON TO THE NEXT PAGE

22. The lines with equations $x = 1$, $x = 6$, $y = -1$, $y = 3$ form a rectangle. What is the area of this rectangle?

(A) 10
(B) 12
(C) 15
(D) 18
(E) 20

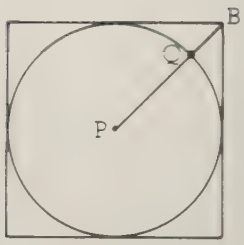
23. A class of 35 students took a test that was scored from 0 to 100. Exactly 20 students received scores greater than or equal to 70. If A is the class average score, which of the following is the LOWEST possible value of A ?

(A) 20 (B) 40 (C) 50 (D) 70 (E) 80

24. A store which formerly sold peppers at 3 pounds for \$1.00 changed the price to 2 pounds for \$0.75. What is the per cent increase in the price per pound?

(A) 25% (B) 20% (C) $16\frac{2}{3}\%$

(D) $12\frac{1}{2}\%$ (E) $6\frac{2}{3}\%$



25. In the square above, Q is the intersection of segment PB and the circle with center P. What is the ratio of the length of QB to the length of PQ ?

(A) $\sqrt{2} - 1$

(B) $\frac{\sqrt{2}}{2}$

(C) $\sqrt{2}$

(D) $\sqrt{2} + 1$

(E) $2\sqrt{2}$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 8B210

VERBAL		MATHEMATICAL	
Section 1	Section 4	Section 2	Section 5
1. D	1. C	1. E	1. B
2. A	2. B	2. D	2. D
3. D	3. E	3. C	3. C
4. A	4. A	4. C	4. B
5. E	5. D	5. E	5. A
6. C	6. B	6. B	6. C
7. B	7. E	7. E	7. E
8. A	8. D	*8. C	8. B
9. D	9. B	*9. D	9. C
10. C	10. C	*10. C	10. A
11. B	11. A	*11. D	11. E
12. E	12. B	*12. A	12. E
13. D	13. D	*13. B	13. B
14. C	14. B	*14. B	14. B
15. B	15. A	*15. A	15. D
16. E	16. B	*16. C	16. C
17. C	17. E	*17. A	17. E
18. A	18. E	*18. A	18. D
19. B	19. C	*19. C	19. A
20. E	20. E	*20. D	20. B
21. D	21. A	*21. A	21. B
22. A	22. B	*22. D	22. E
23. E	23. D	*23. A	23. B
24. A	24. B	*24. B	24. D
25. A	25. E	*25. B	25. A
26. C	26. E	*26. C	
27. C	27. D	*27. D	
28. B	28. C	28. B	
29. D	29. D	29. B	
30. E	30. D	30. D	
31. E	31. B	31. A	
32. B	32. A	32. D	
33. A	33. B	33. A	
34. C	34. A	34. C	
35. B	35. C	35. E	
36. D	36. A		
37. A	37. A		
38. C	38. E		
39. A	39. C		
40. B	40. E		
	41. D		
	42. A		
	43. E		
	44. B		
	45. A		

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. For example, the SAT-verbal test has 85 questions with five answer choices each, and if a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty between different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing ATP scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the form code you marked in item 3 on your answer sheet is the same as the one printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the opposite page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on the back cover.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for section 5 and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on the back cover.

SAT-Verbal Worksheet

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 4: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. Reported SAT-verbal score (See the conversion table on the back cover.) [ ] SAT-Verbal

SAT-Mathematical Worksheet

A. Section 2: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal A

B. Section 2: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. Reported SAT-mathematical score (See the conversion table on the back cover.) [ ] SAT-Math

If you have any questions on these scoring instructions, you may call the phone number below.

If, after following the above scoring directions and checking your work carefully, your results disagree with the verbal or mathematical score reported on your ATP Student Report, you may request a hand scoring of your answer sheet by writing or calling:

College Board ATP
Box 592
Princeton, NJ 08541
Attention: Rescore Request
Telephone: (609) 883-8500

Please indicate whether it is your verbal or mathematical score, or both, that you want to be rescored. When you write, please include a copy of the scoring worksheet in this booklet on which you did your calculations.

Score Conversion Table
Scholastic Aptitude Test
Form Code 8B210

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	470	600
84	790		39	460	590
83	780		38	450	590
82	770		37	450	580
81	760		36	440	570
80	760		35	430	560
79	750		34	430	550
78	740		33	420	540
77	730		32	410	540
76	720		31	410	530
75	720		30	400	520
74	710		29	390	510
73	700		28	390	500
72	690		27	380	490
71	690		26	370	490
70	680		25	370	480
69	670		24	360	470
68	660		23	350	460
67	650		22	350	450
66	650		21	340	440
65	640		20	330	430
64	630		19	330	430
63	620		18	320	420
62	610		17	310	410
61	610		16	310	400
60	600	800	15	300	390
59	590	780	14	290	380
58	590	770	13	290	380
57	580	760	12	280	370
56	570	750	11	270	360
55	570	740	10	270	350
54	560	730	9	260	340
53	550	720	8	250	330
52	550	710	7	250	330
51	540	700	6	240	320
50	530	690	5	230	310
49	530	680	4	220	300
48	520	670	3	220	290
47	510	660	2	210	280
46	510	650	1	200	280
45	500	640	0	200	270
44	490	640	– 1	200	260
43	490	630	– 2	200	250
42	480	620	– 3	200	240
41	470	610	– 4	200	230
			– 5	200	220
			– 6	200	220
			– 7	200	210
			– 8 or below	200	200